

MULTI-LAB TESTING PROCEDURES – UTRE FRAMEWORK

This document defines the multi-laboratory testing procedures for validating UTRE system metrics (Ψ , ΔM , $R(t)$) across independent facilities. The objective is reproducibility, transparency, and convergence across hardware (Illumination Chip, A.L.F.I.E.) and humanware (HRV/EEG coherence trials).

1. Test Site Requirements

- Minimum of three independent laboratories across geographically distributed sites.
- Each lab must meet shielding requirements (Faraday cage or EMI-controlled environment).
- Baseline calibration using reference signal generators and dummy loads.
- Preregistered protocols and IRB approval for human trials.

2. Experimental Design

- Each lab performs identical hardware and/or humanware experiments using shared definitions of Ψ , ΔM , $R(t)$.
- Minimum dataset: 10⁶ timestamped events for hardware trials, 30+ participants per human cohort.
- Sham controls: off-resonance frequency runs, randomized breathing protocols.
- All raw data, spectra, and coherence indices must be logged and time-synchronized.

3. Data Synchronization & Sharing

- Shared repository (Supabase/GitHub) with version-controlled datasets.
- Hashes (SHA256) for all uploads to ensure integrity.
- Metadata must include lab location, device calibration data, environmental logs, and operator notes.

4. Analysis and Replication

- Data from each lab analyzed independently, followed by meta-analysis.
- Effect sizes (Cohen's d , Cliff's δ) computed per lab and compared across sites.
- Convergence criteria: ΔM anomalies reproducible across ≥ 2 labs with $p < 0.05$ (Holm/BH corrected).

5. Reporting & Governance

- All results published with open-access preprints.
- UTRE Oversight Board reviews and certifies convergence events.
- Discrepancies trigger replication or design review.

6. Deliverables

- Raw time-series data (hardware + humanware).
- PSD curves, coherence indices, ΔM trajectories.

- Annotated analysis scripts and reproducibility notebooks.
- Final cross-lab meta-analysis report.